

Collective Seed Storage

Urban Citizens as Active Agents of Agricultural Resilience

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The Need for Seed Diversity

- Over 50,000 edible plants exist worldwide, but **just 15 provide 90% of food energy intake** (e.g., rice, corn, wheat).
- **Genetic diversity is vital to adapt to climate change, extreme weather, pests, and diseases.**
- More than 95% of crop **genetic erosion** articles report changes in diversity, with nearly 80% showing evidence of loss [\[1\]](#).
- **Crop Wild Relatives (CWRs)** are key to providing genetic diversity that cultivated varieties may lack.

Conservation Strategies: In Situ vs Ex Situ

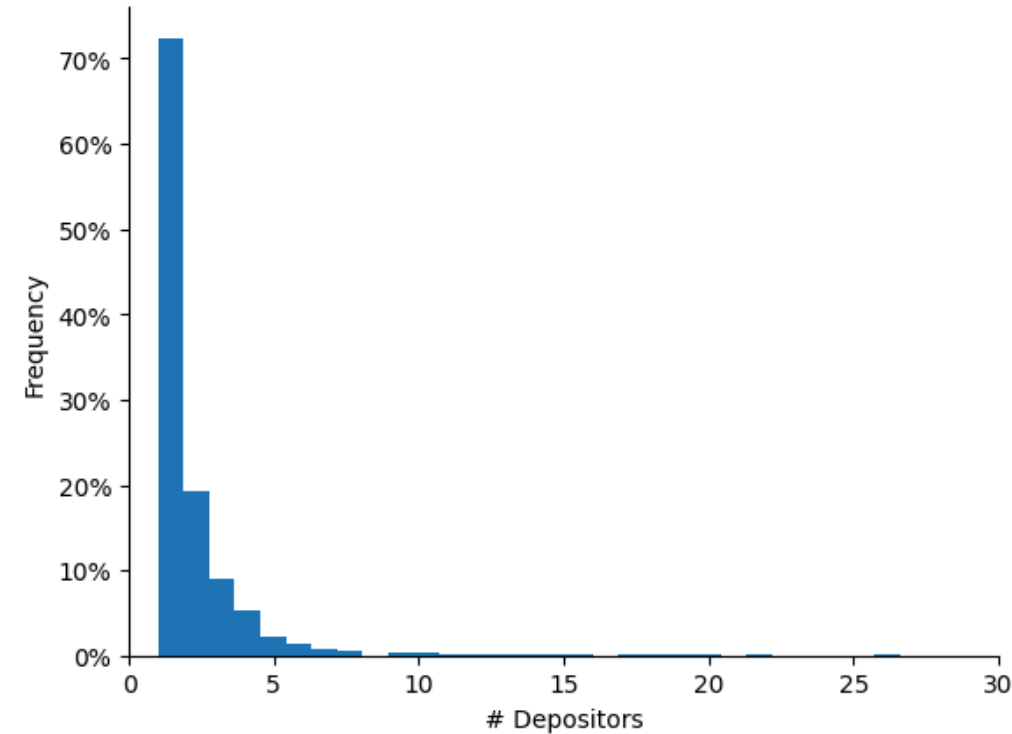
Criteria	In Situ Conservation	Ex Situ Conservation
Location	Natural environment / farms	Seed banks, seed vaults
Evolution	Plants adapt naturally	Static; no natural evolution
Risk of Loss	Higher (climate, land-use)	Lower (controlled conditions)
Access	Local and informal	Global access via formal mechanisms
Cost	Lower short-term cost	Higher cost (infrastructure)
Best Use	Wild relatives, landraces	Major crops, backup

Svalbard Global Seed Vault (SGSV)

- Serves as a global backup facility for seed banks worldwide to ensure long-term survival of agricultural diversity.
- Safeguards over 1.2 million samples representing >6,000 plant species.
- Unanimously considered the **backup seed facility that provides the highest standards of safety and security.**

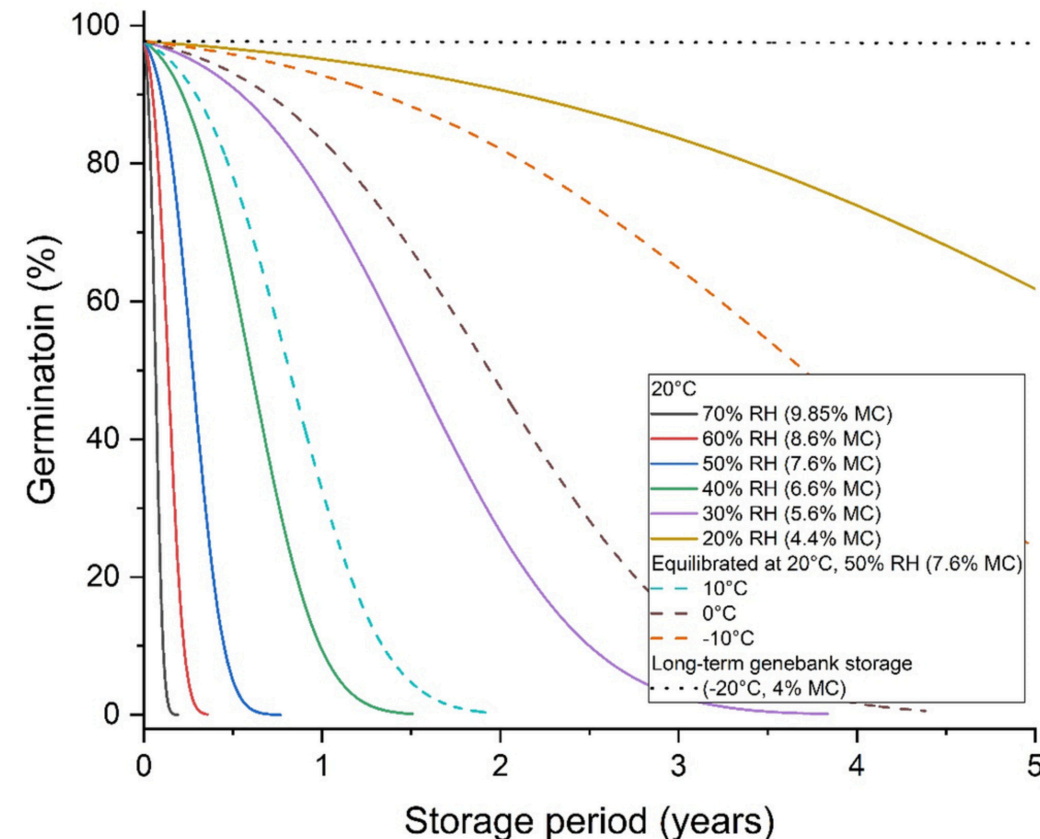
The Open Problem: Limited

- The vast majority of species, mostly CWR in the SGSV, have only a **single depositor** [2].
- **The Threat:** if a disaster hits both the primary seed bank and the vault, the species could be lost.



Orthodox Seeds and Storage

- **Orthodox Seeds:** Approximately 90% of species can retain viability if dried and kept at low temperatures (e.g., wheat, rice, legumes).
- **Storage Standards:** FAO recommends [3] long-term storage at **-18°C and 15% relative humidity**.
- Subzero freezers are acceptable if ideal conditions are not available.



Collective Seed Storage (CSS)

- Domestic freezers provide suitable conditions for seed conservation
- **Vision:** Each household becomes a micro-node in a distributed seed conservation system. Citizens are active contributors to food system resilience and biodiversity conservation
- **Shared Economy Approach:** Sharing a tiny, idle portion of home freezers for seed preservation, to drastically increase redundancy and the overall availability of seeds with minimal infrastructure investments.

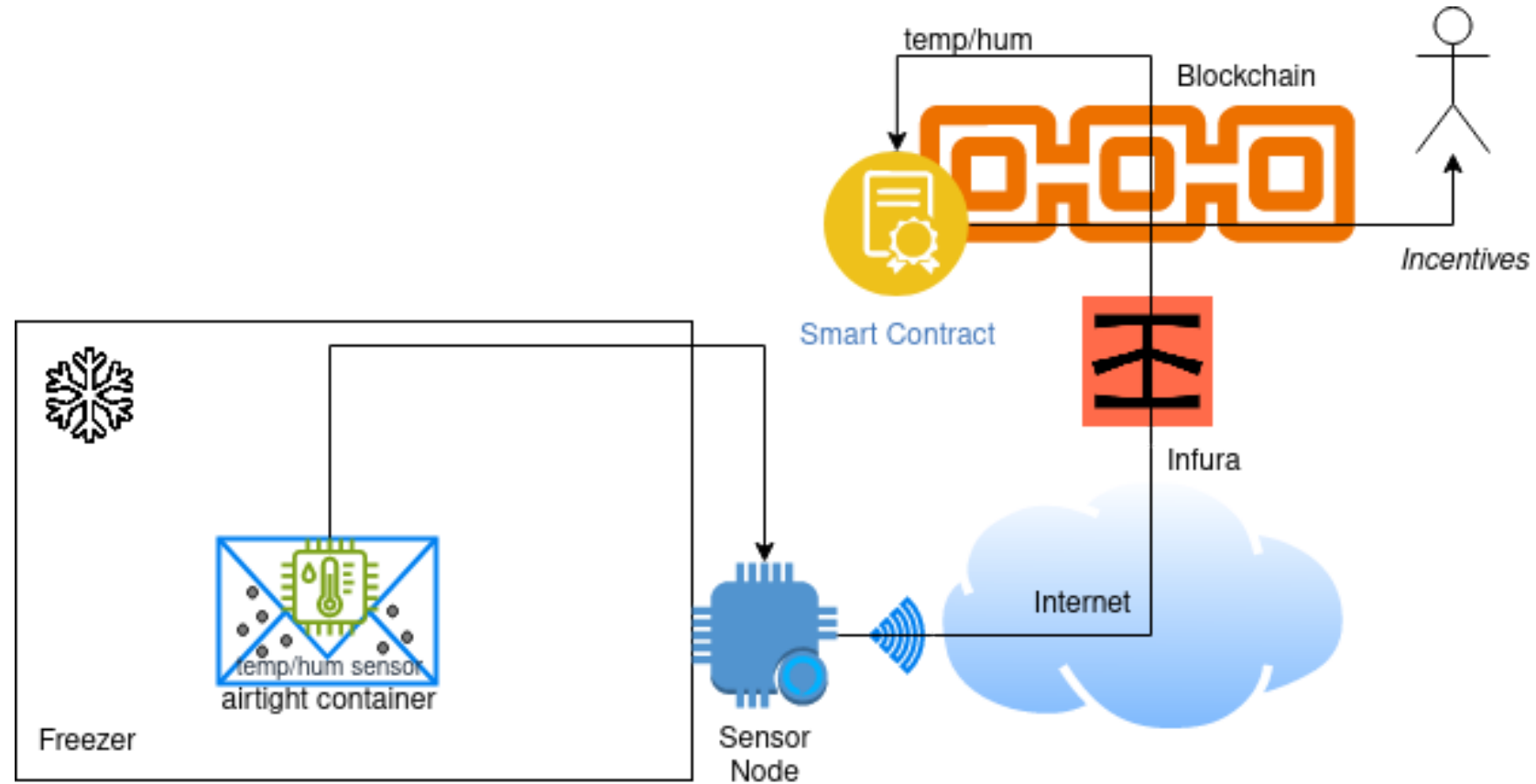


CSS Requirements

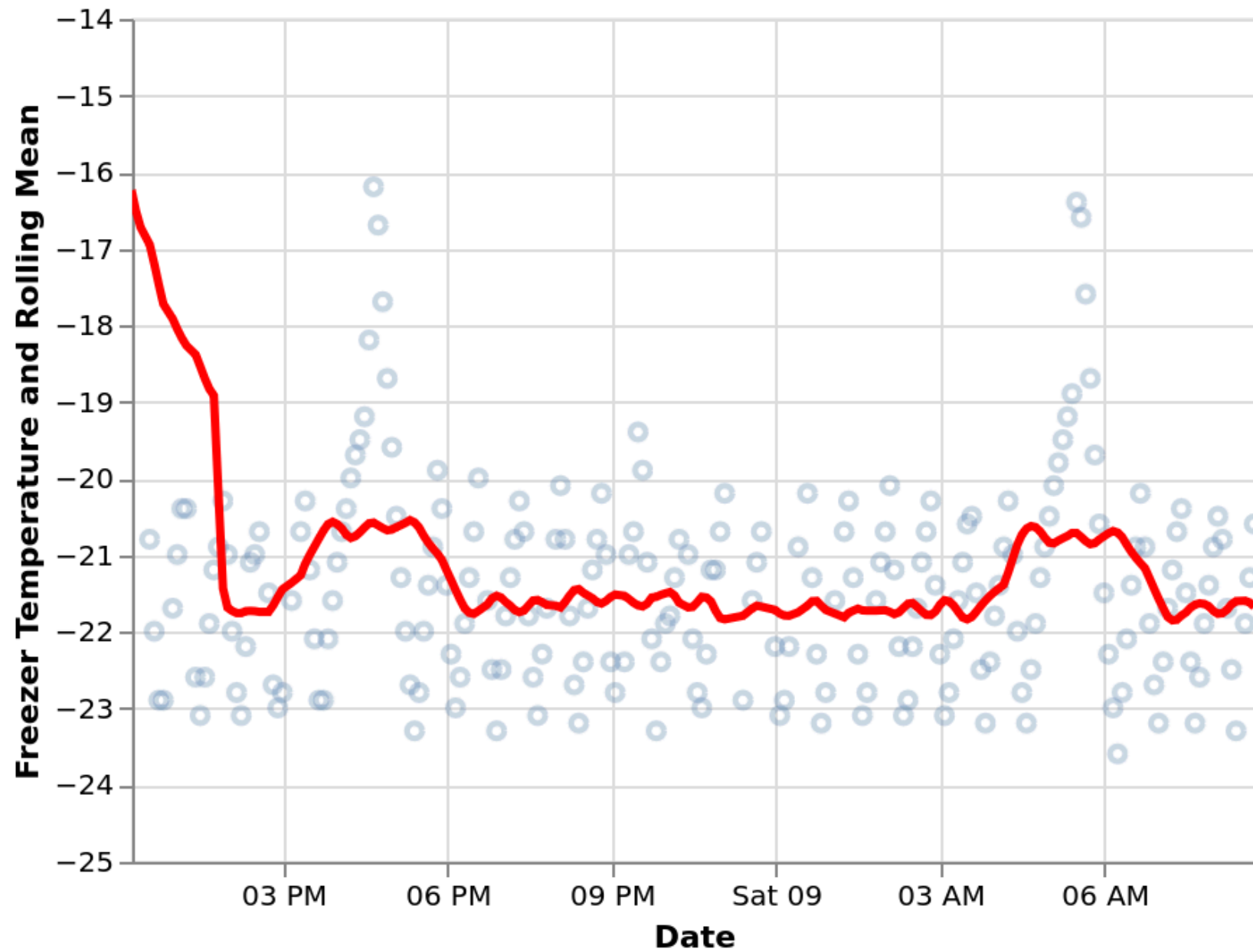
To ensure feasibility and active participation, the system must meet:

1. **Storage Standards (R1):** Must comply with FAO long-term storage conditions (approx. -18°C).
2. **User-Friendly (R2):** Simple installation using off-the-shelf equipment and home WiFi.
3. **No Intrusive Cables (R3):** Minimal wiring to simplify placement in the freezer.
4. **Longevity (R4):** Operate for at least 1 year without user intervention (battery life).
5. **Incentivization (R5):** Implement mechanisms to reward users for their long-term commitment.

Architecture

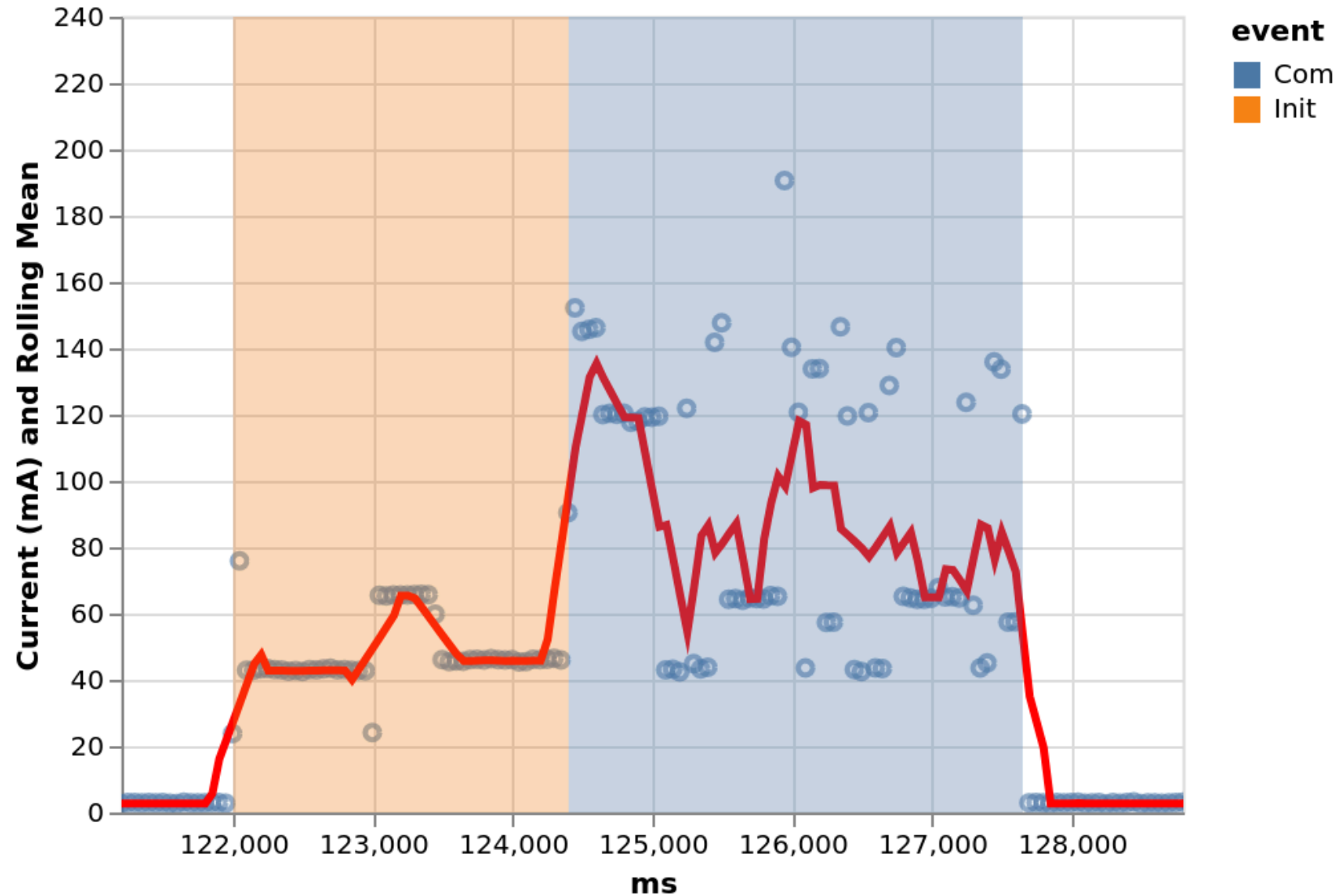


R1: Subzero



R4: lifetime

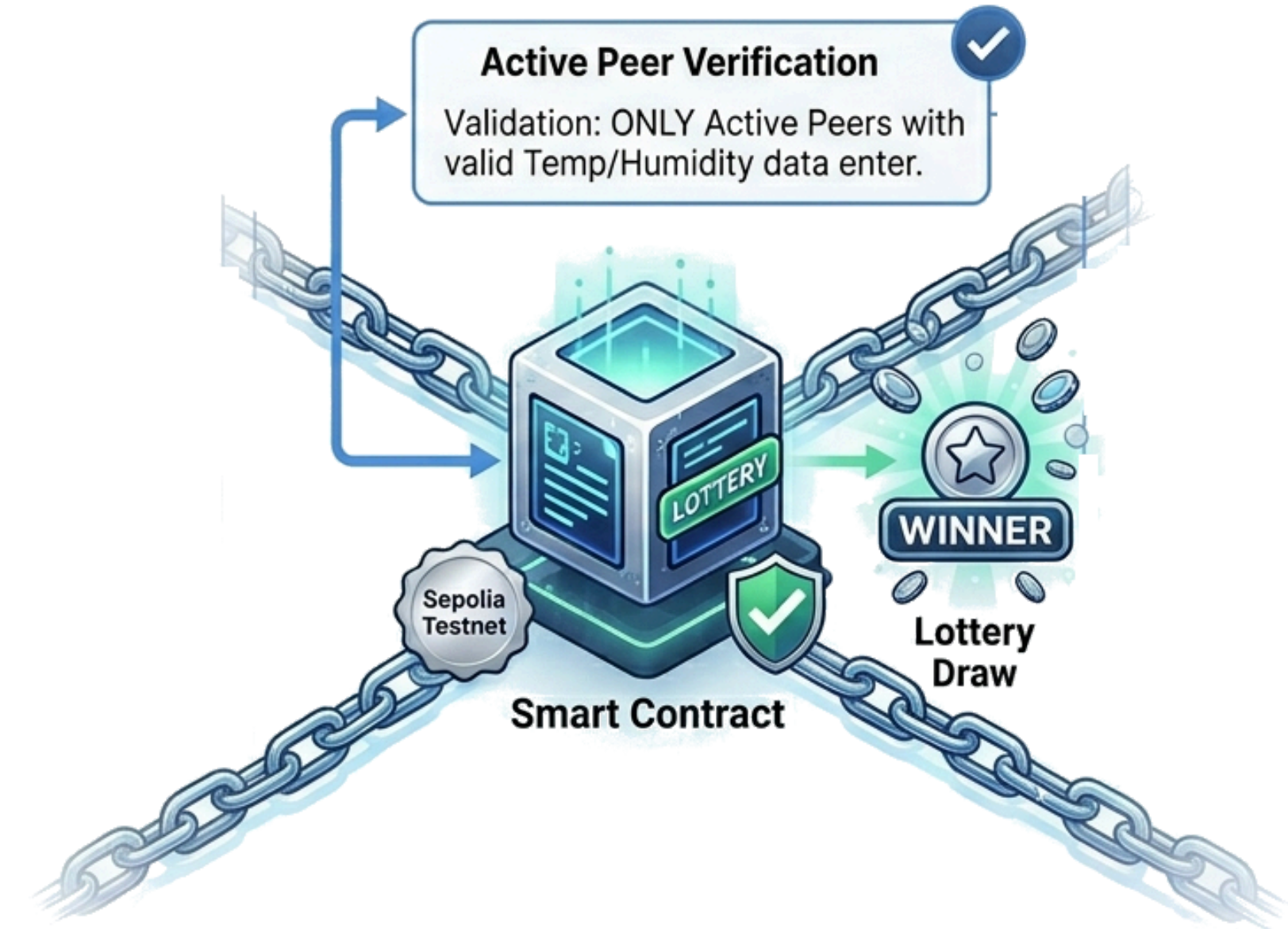
- More than one year
- Duty Cycle: sleep period of >0.73 hours



R5: Incentive

- **The Problem:** Long-term storage (years) might discourage user participation.
- **Solution:** Blockchain Lottery

Smart Contract-Managed Lottery



Conclusions

- CSS can support the preservation of seed diversity by transforming urban citizens into active agents of agricultural resilience.
- households become micro-nodes within a distributed seed conservation system
- Our Proof-of-Concept demonstrates the feasibility of the core technical components underlying this vision.



- A modern approach to seed preservation should promote more active citizen participation (e.g., Seed Guardians) that extends beyond the mere act of storage.
- Although the preservation of seeds in domestic freezers is a novel and valuable practice, it may remain, to some extent, a passive activity.
- CSS can initiate structured knowledge sharing on effective seed preservation.
- Quantitative, objective data can support evidence-based practices and serve as a foundation for more active citizen engagement.

Thank you!

Questions?